Sodium-ion batteries

https://doi.org/10.1038/s41560-026-01994-y

Solvent-enabled fast kinetics

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Electrolyte solvents in sodium-ion batteries have long been regarded as passive ion-transport media, while sluggish Na^+ diffusion in layered cathodes limits kinetics. Now, researchers show that reversible solvent intercalation actively enhances Na^+ transport, enabling fast kinetics with preserved redox chemistry and cycling stability.

Fast charging and discharging while keeping high energy density are key requirements for next-generation sodium-ion batteries (SIBs). However, achieving this combination remains challenging. Although sodium offers advantages in cost and sustainability[1], Na^+ ions diffuse more slowly than Li^+ in most host lattices[2]. In layered oxide cathodes – among the most promising SIB materials[2] – this intrinsic limitation is further exacerbated by lattice distortion, slab gliding[3] and Na–vacancy ordering[4], which collectively impose severe kinetic penalties under high-rate operation.

To address these challenges, most prior strategies have focused on modifying the cathode lattice itself. Approaches such as compositional tuning, defect engineering and particle downsizing have been used to suppress detrimental phase transitions or reduce diffusion barriers[5]. By contrast, the electrolyte has traditionally been viewed as a passive component, serving only to solvate Na^+ ions and transport them to the electrode surface. Solvent molecules, in particular, have been assumed to remain outside layered oxide lattices, with solvent intercalation widely viewed as irreversible or structurally destructive.

Now, writing in *Nature Energy*, Yang Ren, Xiao Cheng Zeng, Qi Liu and colleagues across China and France overturn this long-standing assumption by demonstrating a solvent-enabled kinetic mechanism in a layered sodium manganese oxide cathode[6].

The research team selected glyme-based solvents to promote solvent intercalation in the cathode owing to their linear coordinating geometry, which is favourable for insertion into interlayer galleries. They show that diglyme molecules can reversibly intercalate into a P2-type $Na_{0.67}MnO_2$ (NMO) cathode at high voltages, directly participating in the charge–discharge process. Rather than degrading the host lattice, solvent intercalation stabilizes the layered framework and markedly accelerates Na^+ transport.

When charged to high voltages, solvent molecules intercalate into alternating interlayer spaces of the NMO cathode, accelerating the Na^+ diffusion and stabilizing layered structures (Fig. 1a). During subsequent discharge, the expanded interlayer spacing facilitates fast Na^+ reinsertion. As sodiation proceeds further, diglyme molecules are reversibly extracted from the NMO lattice, thereby releasing active sites for continued Na^+ intercalation (Fig. 1b).

The team further revealed that solvent intercalation occurs selectively at high states of charge where conventional Na^+ diffusion becomes kinetically hindered, rather than at lower states. Their experiments show that the layers of the material expand during charging and return to their original spacing during discharge, demonstrating that the process is fully reversible. Complementary microscopy analyses reveal an accordion-like particle morphology after cycling,

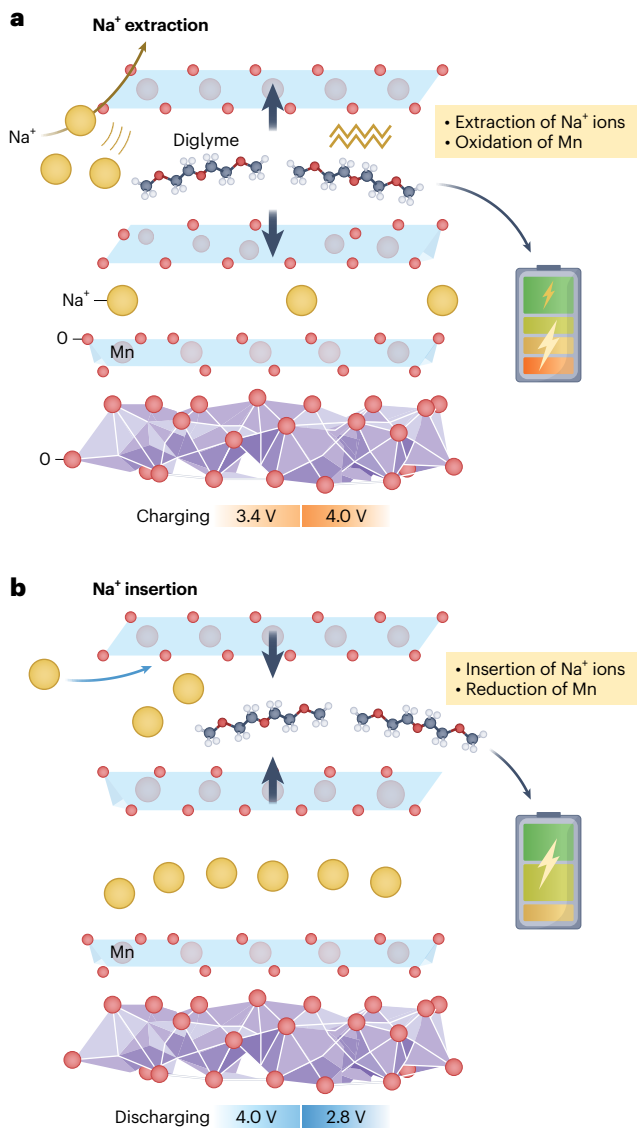

Fig. 1 | Reversible solvent-enabled (de)intercalation process in the layered cathode. a, At high voltages, solvent molecules reversibly intercalate into alternating interlayers of the $Na_{0.67}MnO_2$ cathode, expanding the slab spacing to facilitate Na^+ diffusion and stabilize the layered framework. **b**, Upon discharge, Na^+ rapidly reinserts as solvent molecules are reversibly expelled, restoring the original interlayer configuration and active intercalation sites. Adapted from ref. 6, Springer Nature Ltd.

indicative of controlled lattice breathing rather than structural degradation or mechanical failure.

We envisage that solvent intercalation introduces a dual stabilizing effect. Structurally, solvent molecules act as molecular pillars that prop open MnO_2 slabs, suppressing slab collapse and mitigating destructive phase transitions during deep desodiation. Chemically, strong Na^+–solvent coordination weakens Na–O bonding within the lattice, effectively flattening the energy landscape for in-plane Na^+ migration.

Ren, Zeng, Liu and colleagues performed ab initio density functional theory calculations to provide strong support for the dual stabilizing effect. Compared with the pristine structure, the solvent-intercalated lattice exhibits markedly reduced Na^+ diffusion barriers. In certain configurations, Na^+ ions migrate cooperatively with solvent molecules as transient complexes, further enhancing ionic mobility. Importantly, the research team showed that this acceleration is not confined to the intercalated layers: solvent insertion in one gallery exerts a non-local effect, activating Na^+ transport in adjacent layers as well.

Crucially, the enhancement in kinetics does not come at the expense of altered or degraded redox chemistry. Operando X-ray absorption spectroscopy reveals that manganese redox reactions remain well defined and fully reversible even at high charging rates. The consistent changes observed over a wide range of charging conditions indicate that energy storage occurs throughout the material, rather than being limited to reactions at the surface. This means that allowing solvent molecules to enter the material helps ions move more quickly, while preserving the intrinsic redox behaviour of the cathode.

Ren, Zeng, Liu and colleagues further demonstrated the practical relevance of solvent-enabled fast kinetics in full-cell configurations. By pairing the solvent-intercalating NMO cathode with a graphite anode capable of accommodating solvated Na^+ species, the researchers constructed a full-cell SIB in which solvent molecules participate at both electrodes. This architecture mitigates kinetic bottlenecks across the entire cell, enabling fast charge–discharge operation while maintaining encouraging cycling stability.

Beyond SIBs, this work of Ren, Zeng, Liu and colleagues carries broader implications for electrochemical energy storage. It challenges the long-standing view that electrode materials are 'active' while electrolytes are 'passive', showing instead that designing electrodes and electrolytes together can be powerful. By deliberately enabling solvent participation in bulk electrode processes, their strategy unlocks kinetic pathways that are otherwise inaccessible through host-lattice engineering alone.

Several open questions remain. How general is this solvent-enabled kinetic mechanism across different layered oxides and solvent chemistries? What are the long-term mechanical consequences of repeated solvent-driven lattice breathing under practical cycling conditions? And can similar strategies be extended to other emerging battery systems, including monovalent- or multivalent-ion chemistries, where sluggish diffusion poses an even greater challenge? Addressing these questions will be essential for translating this elegant concept into robust, scalable technologies.

Achieving fast charge–discharge kinetics in layered cathodes without sacrificing structural and redox stability remains one of battery science's most persistent challenges. By demonstrating that such kinetics can be actively enabled through solvent participation, Ren, Zeng, Liu and colleagues offer a fresh perspective on this long-standing problem. Their work suggests that achieving fast charge–discharge kinetics may lie not only in redesigning the electrode[5], but also in rethinking the fundamental role of the electrolyte[6] — not merely as a passive ionic conductor, but as an active participant that can enter the electrode lattice, stabilize its structure, and directly modulate ion-transport kinetics.

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Published online: 06 March 2026

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Competing interests

The author declares no competing interests.